Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3331-WE01

Title: MEng Engineering

Semiconductor Physics and Devices

Time Allowed:	2 hours	
Additional Material provided:	Generic physical constants:- $q = e = 1.60 \times 10^{-19}$ C 1 eV = 1.60×10^{-19} J $m_e = 9.1 \times 10^{-31}$ kg $h = 6.63 \times 10^{-34}$ J s $c = 3.00 \times 10^{8}$ m s^{-1} $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} = 8.85×10^{-14} F cm^{-1} $k = k_B = 1.38 \times 10^{-23}$ J K^{-1} = 8.6×10^{-5} eV K^{-1} $N_A = 6.02 \times 10^{23}$ $mole^{-1}$	
Materials Permitted:	Open books / open notes.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	
	All relevant working must be shown. Only authorised calculators should be used.

Revision:	

Question 1

A thin film of gallium antimonide (GaSb) is doped with acceptors only such that the Fermi level is exactly half way between the intrinsic Fermi level and the valence band maximum, at $T = 300\ K$. *Data for GaSb is provided at the end of the question.*

(a) (i) Determine the type (*n*, *p* or *intrinsic*) of the semiconductor.

(ii) Calculate the density of majority and minority carriers.

[20%]

The film is now illuminated from the top, as shown in **Figure Q1**, with a focused beam that generates a steady-state excess of electrons at $x = 0$ of $\delta n = 10^{12}\ cm^{-3}$. The excess carriers are uniformly distributed in the plane normal to the x axis. The film thickness is $1\ \mu m$ and its cross-section area is $0.01\ cm^2$. Use these steady state conditions for sections (b) and (c).

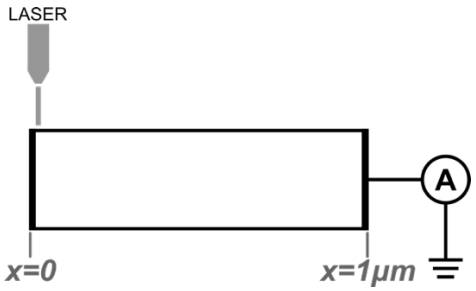


Figure Q1: Cross section of a thin GaSb film and its measurement set-up

(b) (i) Show that the film is much thinner than the electron diffusion length L_n.

(ii) Given your findings in section (b)(i), the recombination term of the continuity equation can be neglected. Derive an expression for the excess of carriers in the x direction of the film $\Delta n(x)$, if the electrode at $x = 1\ \mu m$ is measuring a constant current of $-0.6\ mA$.

[40%]

(c) (i) Explain the difference between steady-state conditions and equilibrium.

Is the semiconductor film in equilibrium, under the conditions set in section (b)?

(ii) Calculate the values of the quasi-Fermi level for electrons with respect to the intrinsic Fermi level, $F_n - E_i$, at $x = 0$ and $x = 1\ \mu m$.

(iii) Draw a schematic sketch of the band diagram of the semiconductor, noting the valence and conduction band edges, the electron and hole quasi-Fermi levels and the intrinsic Fermi level. Focus on a qualitative depiction rather than numerical accuracy.

continued

If you did not solve section (b) you may use:

$p_0 = 10^{16}\ cm^{-3}$, $n_0 = 5 \times 10^{8}\ cm^{-3}$, $\Delta n(0) = 10^{13}\ cm^{-3}$, $\Delta n(1\ \mu m) = 10^{11}\ cm^{-3}$.

[40%]

Data for GaSb:

$$E_g = 0.72\ eV \quad m_n^* = 0.04\ m_e \quad m_p^* = 0.8\ m_e \quad \mu_n = 3{,}000\ \frac{cm^2}{V \cdot s}$$

$$\mu_p = 1{,}000\ \frac{cm^2}{V \cdot s} \qquad \tau_n = 2 \times 10^{-7}\ s \qquad \tau_p = 5 \times 10^{-8}\ s$$

Question 2

(a) (i) Explain what a direct band gap is. Will a material with a direct band gap be efficient both as a light emitting device and a light sensitive device?

(ii) Gallium arsenide (GaAs) has a direct band gap. Will a photodetector built from GaAs be sensitive for (1) fibre optics communications wavelength: $\lambda_{IR} = 1.55\ \mu m$; (2) visible green wavelength: $\lambda_{green} = 550\ nm$; (3) short UV (UV-C) wavelength: $\lambda_{UV} = 200\ nm$?

Data for GaAs is provided at the end of the question.

[12%]

The band structure diagram in **Figure Q2** is that of a GaAs-based pn junction diode in equilibrium, with a surface area of $1\ cm^2$. Its built-in potential is $V_{bi} = 1.23\ V$.

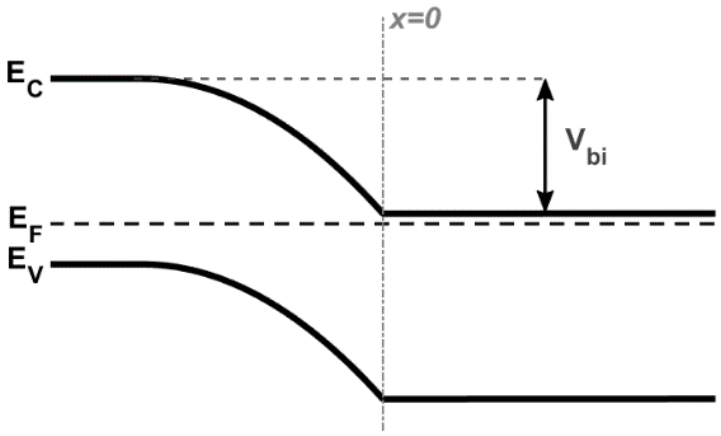


Figure Q2: Band structure of a GaAs pn junction diode

(b) In reverse bias of $V_r = 1.3\ V$, the measured junction capacitance is $C_r = 3.3\ nF$. Calculate the doping in both sides of the junction.

[28%]

(c) The light doping in one side of the junction means that the series resistance of the quasi-neutral region cannot be ignored. If you did not solve section (2), you may use $N_A = 5 \times 10^{14}\ cm^{-3}$.

(i) Calculate the resistance of the lightly doped quasi-neutral region if its length is $2\ cm$.

(ii) Calculate the saturation current of the diode.

(iii) The diode is connected to a current source supplying a bias of $10mA$. What is the potential drop over the entire device?

[48%]

(d) How would you change the diode's doping profile to transform it into a Zener diode? Draw the modified band diagram and explain which parameters have changed to allow tunnelling current.

[12%]

continued

Data for GaAs:

$$E_g = 1.42\ eV \quad \epsilon_r = 12.9 \quad n_i = 2 \times 10^6\ cm^{-3} \quad \mu_n = 8{,}500\ \frac{cm^2}{V \cdot s}$$

$$\mu_p = 400\ \frac{cm^2}{V \cdot s} \qquad \tau_n = 10^{-9}\ s \qquad \tau_p = 5 \times 10^{-11}\ s$$

continued

Question 3

(a) *Fig.Q3 shows* the energy band diagram of a p-type silicon metal oxide semiconductor (MOS) capacitor under flatband conditions.

(i) Sketch a fully annotated flatband energy band diagram for an n-type substrate. Define all the annotations.

(ii) For an n-type substrate, draw and explain very briefly the energy band diagrams in accumulation, depletion and inversion regimes, clarifying the biaising conditions.

[40%]

(b) Consider the ideal n-type silicon MOS capacitor (nSi) at room temperature. The modified workfunction, modified electron affinity and potential difference between the intrinsic Fermi level (E_{Fi}) and the Fermi level (E_F) are $\phi'_m = 3.20V$, $\chi' = 3.25\ V$ and ϕ_{fn}, respectively. The nSi substrate has an energy bandgap of $E_g = 1.12\ eV$ and a donor impurities concentration of $N_d = 10^{16}\ cm^{-3}$. The intrinsic carrier concentration is given by $n_i = 1.5 \times 10^{10}\ cm^{-3}$.

(i) Calculate the metal-semiconductor junction workfunction difference ϕ_{ms}.

(ii) The saturation regime capacitance of the nSi MOS device operated at high frequency is 115 pF. The capacitor has an area is $A = 1\ mm^2$ and the relative permittivity of Si and SiO_2 are $\varepsilon_{rs} = 11.9$ and $\varepsilon_{ox} = 3.9$, respectively. Calculate the oxide thickness d_{ox} of the MOS capacitor and the maximum depletion $W_m(nSi) = \sqrt{\frac{4\varepsilon_s kT}{q^2 N_D} \ln\left(\frac{N_D}{n_i}\right)}$.

(iii) Calculate the minimum capacitance and sketch the current-voltage (C-V) characteristic of the MOS capacitor if the flatband voltage is $V_{FB} = 1.0\ V$.

(iv) Discuss very briefly with illustration how the presence of interface traps will affect the CV curve and the flatband voltage.

[60%]

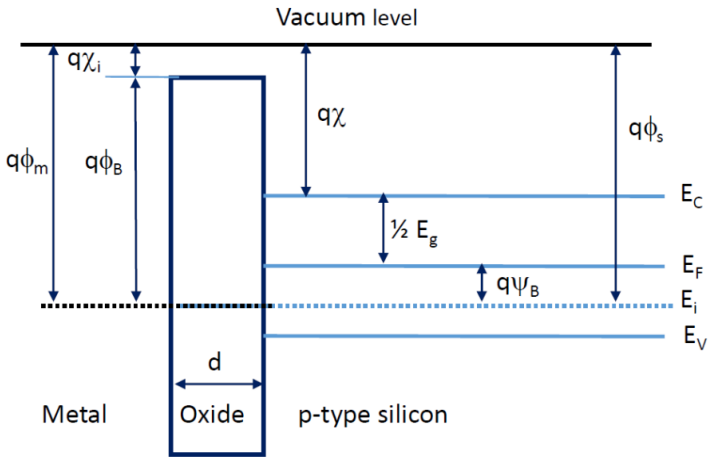


Fig.Q3: Flatband energy band diagram of a p-type silicon MOS capacitor.

continued

Question 4

(a) A memory device is designed using a p-type silicon (pSi) capacitor and gold nanoparticles (AuNPs) as the charge storage elements sandwiched between two identical SiO_2 layers. The capacitance of the capacitive memory device in the accumulation regime is 0.32 nF. The surface area of the capacitor is $1\ mm^2$.

(i) Sketch the structure of the capacitive memory device.

(ii) Calculate the oxide matrix (oxide-AuNPs-oxide) thickness when zero bias is applied, assuming that the permittivity of the matrix is $\varepsilon_{ox} = 3.9$.

(iii) Determine the value of the capacitance in the deep inversion regime assuming that the maximum depletion width is $0.1\ \mu m$. The relative permittivity of Si is $\varepsilon_{rs} = 11.9$.

[32%]

(b) The device exhibits a clockwise hysteresis current voltage characteristics CV curve where the memory windows is characterised by the flatband voltages $V_{FB1} = -1.0\ V$ and $V_{FB2} = +3.0\ V$, respectively.

(i) Sketch and annotate the capacitance-voltage (CV) characteristics of the capacitive memory device.

(ii) Calculate the charge density stored in the memory device using the data above. You may use C_{min}= 200 pF if you have not determined the correct value in (Q4.a.iii).

[32%]

(c) You are asked to integrate a MOS capacitor memory device ($100\ \mu m^2$ area) with a MOSFET to form a dynamic random access memory (DRAM) cell. It is assumed that a monolayer of AuNPs is imprinted on the pSi MOS capacitor to act as charge storage nodes.

(i) Sketch and annotate the structure of a DRAM device. What is meant by oxidation and metallisation in the context of a DRAM cell?

(ii) Propose a realistic method for the fabrication of the capacitive memory device as part of an array of 10 DRAM cells, highlighting only the key steps of the pattern definition, metallisation and integration with the DRAM. Ignore the fabrication of the MOSFET.

(iii) What variant of conventional photolithography will be needed when the capacitor area shrinks down submicron sizes? Justify your answer and show the supporting illustrations.

[36%]

END